

How Forces Affect Motion

Force, net force, and Newton's three laws — the framework for predicting whether an object stays at rest, moves at constant velocity, or accelerates.

01 Overview

This guide explores the underlying causes of changes in the motion of objects, moving beyond simple descriptions of position and velocity. It details the nature of force as a physical quantity and examines how objects respond to various interactions through the framework of Newton's three laws of motion. By mastering these concepts, one can predict whether an object will remain at rest, move at a constant velocity, or accelerate based on the net forces acting upon it.

02 Key Concepts & Definitions

Force — A physical quantity that can cause an object to move from rest, change its speed or direction, or alter its shape (e.g., squeezing a lemon). It is a vector quantity, requiring both magnitude and direction.

Newton (N) — The SI unit of force. The full name is written in lowercase ("newton"), while the symbol is capitalized ("N").

Net Force — The combined effect of all forces acting on an object. If forces act in the same direction, the net force is the **sum** of their magnitudes. If forces act in opposite directions, the net force is the **difference** between their magnitudes, acting in the direction of the larger force.

Balanced Forces — Forces acting on an object that are equal in magnitude and opposite in direction. They result in a net force of zero and do not change the object's state of

motion.

Unbalanced Forces — Forces that result in a non-zero net force, causing a change in the object's velocity (acceleration).

Friction — A force that arises between surfaces in contact, acting in the direction opposite to the motion (or intended motion) of the object.

Normal Force — An upward force exerted by a surface perpendicular to the object placed on it, typically balancing the object's weight.

Inertia — The inherent tendency of an object to resist any change in its state of rest or uniform motion.

Momentum — The product of an object's mass and its velocity ($p = mv$). The direction of momentum is the same as the velocity.

03 Formulas & Derivations

Newton's Second Law of Motion

The fundamental relationship between force, mass, and acceleration.

$$F = ma$$

Variable	Definition	Unit
F	Net force	Newtons (N)
m	Mass of the object	Kilograms (kg)
a	Acceleration	Meters per second squared (m/s^2)

Usage: Used to calculate the force required to produce a specific acceleration in a known mass, or to determine the resulting acceleration when a force is applied.

Derivation: Observations from experiments (like the cart-and-pulley activity) show that acceleration (a) is directly proportional to the net force (F) and inversely proportional to the mass (m):

$$a \propto \frac{F}{m}$$

Combining these gives $F = ma$.

Weight (Gravitational Force)

The force with which the Earth pulls an object toward itself.

$$F = mg$$

Variable	Definition	Value / Unit
g	Acceleration due to gravity	9.8 m/s ² (or 10 m/s ² for estimates)

Force in Terms of Momentum

A more complete form of the Second Law, applicable even when mass is not constant.

The rate of change of momentum is proportional to the net force and acts in the direction of the force:

$$F \propto \frac{\Delta p}{\Delta t}$$

04 Worked Examples

Example 1 — Calculating Net Force

Two forces of 10 N and 6 N act on a block in opposite directions (10 N to the right, 6 N to the left). Calculate the magnitude and direction of the net force.

1. Identify the directions. Since they are opposite, subtract the magnitudes.
2. $10 \text{ N} - 6 \text{ N} = 4 \text{ N}$.

Result: net force = 4 N, acting to the right (direction of the larger force).

Example 2 — Constant Velocity and Newton's First Law

A person exerts a forward force on a moving box that is exactly equal to the force of friction (5 N applied, 5 N friction). Will the box stop?

1. The applied force and friction are equal and opposite.
2. Net force = 5 N – 5 N = 0 N.

Conclusion: by Newton's First Law, since net force is zero, the box continues moving at constant velocity – it will not come to rest unless an additional net force acts on it.

Example 3 — Force Required for a Weightlifter

A weightlifter holds a barbell steady. The barbell consists of a 10 kg bar and two 10 kg weights (one on each side). Calculate the force the weightlifter must apply.

1. Total mass: 10 kg + 10 kg + 10 kg = 30 kg.
2. Use the gravitational force formula, $F = mg$, with $g = 9.8 \text{ m/s}^2$.
3. $F = 30 \text{ kg} \times 9.8 \text{ m/s}^2 = 294 \text{ N}$.

Conclusion: to keep the barbell steady (balanced forces), the weightlifter must apply an upward force of 294 N.

05 Common Misconceptions

MYTH

A force is needed to keep an object in motion.

FACT

According to Galileo and Newton's First Law, an object will continue moving indefinitely at a constant velocity if there is no net force (like friction) to stop it. We only apply "continuous force" in daily life to *counteract* friction.

MYTH

Balanced forces mean an object must be at rest.

FACT

Balanced forces (net force = 0) mean there is no *change* in motion. An object can be either at rest or moving at a constant velocity in a straight line.

MYTH

Friction only depends on the weight of the object.

FACT

Friction also depends heavily on the nature of the surfaces in contact — a stack of coins travels farther on polished marble than on a wooden table.

06 Practice Questions

Questions

► SHOW ANSWER KEY

1. Identify the SI unit of force and its symbol.
2. State Newton's First Law of Motion in terms of "net force."
3. If two people push a stalled car in the same direction with forces of 40 N and 60 N, what is the net force?
4. Define "Inertia" and explain its relationship to Newton's First Law.
5. A 5 kg mass is accelerating at 2 m/s^2 . Calculate the net force acting on it.
6. What is the force of gravity (weight) acting on an object with a mass of 2 kg? (Use $g = 9.8 \text{ m/s}^2$.)
7. If the force applied to an object is doubled while the mass remains constant, how does the acceleration change?
8. Describe the motion of an object when the net force acting on it is zero.
9. Why does a fielder pull their hands backward while catching a fast-moving cricket ball?
10. A box is being pushed on a floor. List the four forces typically acting on it.

07 Quick-Reference Summary

Newton's Three Laws

- 1 **First Law:** No net force = no change in motion (constant velocity or rest).
- 2 **Second Law:** $F = ma$. Acceleration is in the direction of the net force.
- 3 **Third Law:** Illustrated by the canoe example — pushing water backward moves the canoe forward (action / reaction).

Core Formulas

Concept	Formula	Notes
Net Force	$F_{\text{net}} = \Sigma F$	Add if same direction; subtract if opposite.
Force	$F = ma$	SI unit: newton (N).
Weight	$W = mg$	$g \approx 9.8 \text{ m/s}^2$.
Momentum	$p = mv$	Units: kg·m/s.

Key Reminders

- **Direction matters:** force is a vector. If you change direction, you change the force's effect.
- **Friction** always opposes motion. Smoother surfaces mean less friction and a longer travel distance.
- **Measurement:** use a spring balance to measure force magnitude.
- **Scale of force:** a light touch is roughly 1 millinewton (10^{-3} N). Specialized experiments can measure down to yoctonewtons (10^{-24} N).

